

White paper: notation and terminology across the compendium

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A re-audit of mathematical notation and controlled vocabulary in the manuscripts under analysis/report/, after two repair passes. Records what was fixed, what the repairs themselves broke, and what remains.

Method

Sixteen manuscript sources were searched for the symbols carrying the programme's substantive quantities and for the labels naming its designs, architectures, specifications, and studies. The checks are now mechanized as a linter (`notation-lint.pl`) that skips fenced R chunks and `verbatim` blocks, strips `\texttt{}` and backtick spans, and reports violations of the rules in `analysis/report/NOTATION.md`. Every manuscript was re-rendered after each change; counts below are from the sources and from `pdftotext` on the rendered output.

The corpus is smaller than in the first edition. Papers 01 and 02 have since been reorganized: their narrative variants were promoted to `report.Rmd` and the earlier full and slim versions moved to per-paper `archive/` directories. This audit reflects the promoted files.

State after the second pass

The linter reports **clean** for all sixteen manuscripts except one known false positive (a bare `Sx` inside a backtick span that crosses a line break). All sixteen render. The nine inconsistencies of the first edition now stand as follows.

#	Finding (first edition)	State
N1	Drug indicator: D_{it} used for both the binary state and the continuous exposure	Fixed
N2	c_{bm} used for both a correlation and a regression multiplier	Fixed
N3	Half-life quoted in weeks and in days with no conversion	Fixed by units paragraphs
N4	Four symbols for one estimand	Standardized on $\beta_{bm:D}$
N5	Outcome alternates between Y and the code identifier <code>Sx</code>	Fixed in model definitions
N6	Label collisions across architectures, specifications, sweeps, studies	Fixed, after one false start
N7	Two sample-size conventions (per path, total)	Documented, not resolved
N8	Design labels alternate abbreviation and prose	Open (editorial)
N9	Sign convention stated only in one paper	Open (editorial)

What the second pass found that the first pass missed

The specification relabel was incomplete. The first pass renamed the three carryover specifications A1/A2/A3 to M1/M2/M3 in the carryover manuscript and its variants, but four other papers cite those specifications and were not touched. For a day the compendium said M2 in one paper and A2 in four others, which is worse than the original state, because a reader cross-referencing them would conclude they were different objects.

The chosen replacement collided with an existing scheme. The calibration manuscript already uses M0 through M3 for its working-covariance model ladder, and those labels are keys into archived simulation output, not free text. M1/M2/M3 for specifications therefore reintroduced exactly the class of collision the pass existed to remove. The specifications are now **E1/E2/E3**, after E for the exposure regressor that distinguishes them; E is unused anywhere else in the corpus.

The architectures were the deeper problem. A, B, and C are opaque (a reader must memorize which is which) and they are the reason the specification labels collided in the first place. They are now named by the channel that carries the interaction:

Prose	Legends and tables	Data value
the mean-moderation architecture	Mean	<code>mean_moderation</code>
the covariance-moderation architecture	Covariance	<code>mvn</code>
the dual-channel architecture	Dual	<code>combined</code>

This touched 581 label occurrences across 16 files plus the figure drivers. Data values are unchanged, so no simulation was re-run; the figures were regenerated from existing summaries.

A second British-spelling family survived the first pass, because the first pass enumerated word stems rather than patterns: `flavour`, `rigour`, `endeavour`, `labour`, `tumour`. Fixed in four files.

What the repairs themselves broke

Reporting this is the point of a second audit. Three defects were introduced by the first pass and caught here.

Data-side labels were renamed along with display labels. The specification rename reached inside R chunks, changing `method_levels <- c('A1', ...)` and three `filter(spec %in% c('A2', 'A3'))` calls. Factor levels and filters index archived data values; renaming them silently produced empty selections and `NA` rows rather than an error. Fixed by keeping data values at A1/A2/A3 and mapping to E1/E2/E3 at display time through a `spec_display()` helper, with the mapping documented in each paper’s reproducibility section.

A display helper returned NA for unmapped inputs. The first version of `spec_display()` indexed a named vector, so the two robust-inference methods (`lme+CR2`, `GEE+MD`) that are not specifications came back `NA` and printed as `NA` in the development-review tables. Now passes unmapped values through.

Three manuscript files were truncated to zero bytes. A shell loop copied each source to a temporary file, transformed it, and moved the result back. When the cloud file provider returned a spurious `ENOENT` for a source that had just been moved by a concurrent reorganization, the `cp` failed but the shell had already created an empty output file, which `mv` then placed over the source. The files were restored from scratchpad backups. Every subsequent script verifies that the source is non-empty before producing output and that the result has not shrunk before writing back. This is a sharper version of the hazard the project guardrails already warn about: the danger is not only in-place editors, but any pipeline whose failure mode is an empty file.

What remains open

N7, the sample-size convention. One paper reports N per randomization path and others report it in total. The combined-DGP manuscript documents its own instance and defers to a pending boundary re-run. This needs a simulation, not an edit.

N8 and N9 are editorial: design labels still alternate between abbreviation and spelled-out form, and the sign convention is stated in only one paper. Both are best folded into the `rgt` writing pass rather than done mechanically now.

The rgt prose remains unwritten in every paper that still carries the three-part scaffold, so a third terminology pass will be needed once the author’s text exists. That is the argument for keeping the linter: it converts these two one-off passes into a check that can run on every render.

A caution on scope. This audit verifies internal consistency of symbols and labels. It does not verify that any numerical result is correct, and the label changes were deliberately confined to display so that no stored result was touched. Two papers were promoted to narrative variants during the pass; their archived predecessors were not re-audited and will not match this scheme if they are ever revived.